Seminar 1

Today: start from the **theorem**, leave holes `?_`, and **fill them in** — the way you actually type it in the editor.

And after each proof we'll write down **one short rule**. By the end we'll have our own **rulebook**.

Plan

1. `implFlipPremises` — `(A → B → C) → B → A → C`
2. `implS` — `(A → B → C) → (A → B) → A → C`
3. `or_elim_tutorial` — `A ∨ B → (A → C) ∧ (B → C) → C`
4. `iff_trans` — `(A ↔ B) → (B ↔ C) → (A ↔ C)`
5. `iff_and_congr` — `(A ↔ B) → (C ↔ D) → (A ∧ C ↔ B ∧ D)`
6. `implPhi` — `(A → B → C) → (D → A) → (D → B) → D → C`
7. `contrapositive_double` — `(A → B) → ¬¬A → ¬¬B`
8. `implApplyToChain` — `(((P → Q) → Q) → R) → (R → S) → P → S`

The method

For any goal, repeat until there are no holes:

1. **Read** the type of the current hole `?_`.
2. Is it `X → Y`? **Introduce** the argument: `fun x => ?_`.
3. **Deconstruct** what you *have* (`∧` → `⟨a, b⟩`, `∨` → cases).
4. **Build** what you *need* (`∧` / `↔` → `⟨?_, ?_⟩`, `∨` → `Or.inl` / `Or.inr`).
5. **Match** each `?_` to something in scope.

implFlipPremises

```
theorem implFlipPremises {A B C : Prop} :
    (A → B → C) → B → A → C :=
  ?_
```

1 goal

```
A B C : Prop
⊢ (A → B → C) → B → A → C
```

implFlipPremises

```
theorem implFlipPremises {A B C : Prop} :
    (A → B → C) → B → A → C :=
  fun f b a => ?_
```

arguments introduced — goal still open

```
f : A → B → C
b : B
a : A
⊢ C
```

implFlipPremises

```
theorem implFlipPremises {A B C : Prop} :
    (A → B → C) → B → A → C :=
  fun f b a => f a b
```

fill the hole

```
f : A → B → C
b : B ,  a : A
f a b : C     -- f's order: A then B
```

No goals

Rule 1: `→` groups to the right — `X → Y → Z` means `X → (Y → Z)`; each top-level left arrow is one `fun` argument.

implS

```
theorem implS {A B C : Prop} :
    (A → B → C) → (A → B) → A → C :=
  fun f g a => ?_
```

1 goal

```
f : A → B → C
g : A → B
a : A
⊢ C
```

implS

```
theorem implS {A B C : Prop} :
    (A → B → C) → (A → B) → A → C :=
  fun f g a => f a (?_)
```

applied `f a` — the hole must be a `B`

```
f : A → B → C
g : A → B
a : A
f a : B → C   -- so the hole needs a B
⊢ B
```

implS

```
theorem implS {A B C : Prop} :
    (A → B → C) → (A → B) → A → C :=
  fun f g a => f a (g a)
```

filling the hole

```
a   : A          -- for f
g a : B          -- uses a again
f a (g a) : C
```

No goals

Rule 2: a term may be used more than once.

or_elim_tutorial

```
theorem or_elim_tutorial {A B C : Prop} :
    A ∨ B → (A → C) ∧ (B → C) → C :=
  ?_
```

1 goal — just read the type

```
A B C : Prop
⊢ A ∨ B → (A → C) ∧ (B → C) → C
-- have: an  A ∨ B  and an  (A → C) ∧ (B → C)
-- the second is an ∧  →  deconstruct it
```

or_elim_tutorial

```
theorem or_elim_tutorial {A B C : Prop} :
    A ∨ B → (A → C) ∧ (B → C) → C :=
  fun h ⟨fa, fb⟩ => ?_
```

write the arguments — `⟨fa, fb⟩` splits the ∧

```
h  : A ∨ B
fa : A → C
fb : B → C
⊢ C
```

or_elim_tutorial

```
theorem or_elim_tutorial {A B C : Prop} :
    A ∨ B → (A → C) ∧ (B → C) → C :=
  fun h ⟨fa, fb⟩ =>
    match h with
    | Or.inl a => ?_
    | Or.inr b => ?_
```

2 goals — one per constructor

```
Or.inl a:  a : A , fa : A → C , fb : B → C
⊢ C

Or.inr b:  b : B , fa : A → C , fb : B → C
⊢ C
```

or_elim_tutorial

```
theorem or_elim_tutorial {A B C : Prop} :
    A ∨ B → (A → C) ∧ (B → C) → C :=
  fun h ⟨fa, fb⟩ =>
    match h with
    | Or.inl a => fa a
    | Or.inr b => fb b
```

left `fa a`, right `fb b`

```
fa a : C
fb b : C
```

No goals

Rule 3: if you have an ∧ or ∨, deconstruct it (`⟨a,b⟩` / `match`).

iff_trans

```
theorem iff_trans {A B C : Prop} :
    (A ↔ B) → (B ↔ C) → (A ↔ C) :=
  fun hab hbc => ?_
```

arguments introduced — goal still open

```
hab : A ↔ B
hbc : B ↔ C
⊢ A ↔ C
```

iff_trans

```
theorem iff_trans {A B C : Prop} :
    (A ↔ B) → (B ↔ C) → (A ↔ C) :=
  fun hab hbc =>
    ⟨ ?_ ,
      ?_ ⟩
```

an ↔ is a pair — build `⟨ ?_ , ?_ ⟩`

```
hab.mp  : A → B    hab.mpr : B → A
hbc.mp  : B → C    hbc.mpr : C → B
-- goals:
mp   ⊢ A → C
mpr  ⊢ C → A
```

iff_trans

```
theorem iff_trans {A B C : Prop} :
    (A ↔ B) → (B ↔ C) → (A ↔ C) :=
  fun hab hbc =>
    ⟨ fun a => ?_ ,
      fun c => ?_ ⟩
```

each hole is an arrow → introduce `fun … => ?_`

```
hab.mp  : A → B    hab.mpr : B → A
hbc.mp  : B → C    hbc.mpr : C → B
-- goals:
mp   a : A ⊢ C
mpr  c : C ⊢ A
```

iff_trans

```
theorem iff_trans {A B C : Prop} :
    (A ↔ B) → (B ↔ C) → (A ↔ C) :=
  fun hab hbc =>
    ⟨ fun a => hbc.mp (hab.mp a),
      fun c => hab.mpr (hbc.mpr c) ⟩
```

forward direction

```
hab.mp a           : B
hbc.mp (hab.mp a) : C
```

No goals

Rule 4: an ↔ is a pair `⟨mp, mpr⟩`; project with `.mp` / `.mpr`.

iff_and_congr

```
theorem iff_and_congr {A B C D : Prop}
    (hab : A ↔ B) (hcd : C ↔ D) :
    A ∧ C ↔ B ∧ D :=
  ⟨ ?_ ,
    ?_ ⟩
```

an ↔ is a pair — build `⟨ ?_ , ?_ ⟩`

```
hab.mp  : A → B    hab.mpr : B → A
hcd.mp  : C → D    hcd.mpr : D → C
-- goals:
mp   ⊢ A ∧ C → B ∧ D
mpr  ⊢ B ∧ D → A ∧ C
```

iff_and_congr

```
theorem iff_and_congr {A B C D : Prop}
    (hab : A ↔ B) (hcd : C ↔ D) :
    A ∧ C ↔ B ∧ D :=
  ⟨ fun h1 => ?_ ,
    fun h2 => ?_ ⟩
```

each hole is an arrow → introduce `fun h1 / h2 => ?_`

```
hab.mp  : A → B    hab.mpr : B → A
hcd.mp  : C → D    hcd.mpr : D → C
-- goals:
mp   h1 : A ∧ C ⊢ B ∧ D
mpr  h2 : B ∧ D ⊢ A ∧ C
```

iff_and_congr

```
theorem iff_and_congr {A B C D : Prop}
    (hab : A ↔ B) (hcd : C ↔ D) :
    A ∧ C ↔ B ∧ D :=
  ⟨ fun h1 => ⟨ ?_ , ?_ ⟩ ,
    fun h2 => ⟨ ?_ , ?_ ⟩ ⟩
```

each goal is an ∧ — build `⟨ ?_ , ?_ ⟩`

```
h1.left : A , h1.right : C
h2.left : B , h2.right : D
-- goals:
mp   ⊢ B , ⊢ D
mpr  ⊢ A , ⊢ C
```

iff_and_congr

```
theorem iff_and_congr {A B C D : Prop}
    (hab : A ↔ B) (hcd : C ↔ D) :
    A ∧ C ↔ B ∧ D :=
  ⟨ fun h1 => ⟨ hab.mp h1.left , ?_ ⟩ ,
    fun h2 => ⟨ ?_ , ?_ ⟩ ⟩
```

first row, left: `hab.mp h1.left : B`

```
h1.left : A ,  hab.mp h1.left : B
-- remaining:
mp   ⊢ D
mpr  ⊢ A , ⊢ C
```

iff_and_congr

```
theorem iff_and_congr {A B C D : Prop}
    (hab : A ↔ B) (hcd : C ↔ D) :
    A ∧ C ↔ B ∧ D :=
  ⟨ fun h1 => ⟨ hab.mp h1.left , hcd.mp h1.right ⟩ ,
    fun h2 => ⟨ ?_ , ?_ ⟩ ⟩
```

first row done: `⟨ ... ⟩ : B ∧ D`

```
h1.right : C ,  hcd.mp h1.right : D
-- remaining:
mpr  ⊢ A , ⊢ C
```

iff_and_congr

```
theorem iff_and_congr {A B C D : Prop}
    (hab : A ↔ B) (hcd : C ↔ D) :
    A ∧ C ↔ B ∧ D :=
  ⟨ fun h1 => ⟨ hab.mp h1.left , hcd.mp h1.right ⟩ ,
    fun h2 => ⟨ hab.mpr h2.left , hcd.mpr h2.right ⟩ ⟩
```

second row: mirror with `.mpr`

```
h2.left  : B ,  hab.mpr h2.left  : A
h2.right : D ,  hcd.mpr h2.right : C
```

No goals

Rule 5: to BUILD an ∧ / ↔, build each side and combine with `⟨_, _⟩`.

implPhi

```
theorem implPhi {A B C D : Prop} :
    (A → B → C) → (D → A) → (D → B) → D → C :=
  fun f g h d => ?_
```

1 goal

```
f : A → B → C
g : D → A
h : D → B
d : D
⊢ C
```

implPhi

```
theorem implPhi {A B C D : Prop} :
    (A → B → C) → (D → A) → (D → B) → D → C :=
  fun f g h d => ?_
```

what we can derive

```
g d : A     -- g : D → A  applied to d
h d : B     -- h : D → B  applied to d
f needs :  A  then  B
```

implPhi

```
theorem implPhi {A B C D : Prop} :
    (A → B → C) → (D → A) → (D → B) → D → C :=
  fun f g h d => f (g d) (h d)
```

fan-out then fan-in

```
g d : A          -- d feeds g
h d : B          -- d feeds h
f (g d) (h d) : C
```

No goals

contrapositive_double

```
theorem contrapositive_double {A B : Prop} :
    (A → B) → ¬¬A → ¬¬B :=
  ?_
```

read the type — `¬X` is `X → False`

```
¬¬A  =  (A → False) → False
¬¬B  =  (B → False) → False
brackets explicit (→ groups right):
(A → B) → ( ¬¬A → ¬¬B )
```

contrapositive_double — the full type

```
(A → B) → ((A → False) → False) → (B → False) → False
```

¬X is X → False, so ¬¬A = (A → False) → False and ¬¬B = (B → False) → False.

contrapositive_double

```
theorem contrapositive_double {A B : Prop} :
    (A → B) → ¬¬A → ¬¬B :=
  fun f nna nb => ?_
```

extract the three arguments

```
f   : A → B
nna : ¬¬A     -- (A → False) → False
nb  : ¬B      -- B → False
⊢ False
```

contrapositive_double

```
theorem contrapositive_double {A B : Prop} :
    (A → B) → ¬¬A → ¬¬B :=
  fun f nna nb => nna ?_
```

apply `nna` — it needs a `¬A`

```
nna : ¬¬A     -- (A → False) → False
nna wants a  ¬A  (= A → False)
⊢ A → False
```

contrapositive_double

```
theorem contrapositive_double {A B : Prop} :
    (A → B) → ¬¬A → ¬¬B :=
  fun f nna nb => nna (fun a => ?_)
```

introduce `fun a => ?_` (a : A)

```
a  : A
f  : A → B     f a : B
nb : ¬B        -- B → False
⊢ False
```

contrapositive_double

```
theorem contrapositive_double {A B : Prop} :
    (A → B) → ¬¬A → ¬¬B :=
  fun f nna nb => nna (fun a => nb (f a))
```

inside `fun a => ?_` , a : A

```
f a      : B
nb (f a) : False
nna (fun a => nb (f a)) : False
```

No goals

Rule 6: `¬X` is `X → False`; a proof of `False` closes any goal.

implApplyToChain

```
theorem implApplyToChain {P Q R S : Prop} :
    (((P → Q) → Q) → R) → (R → S) → P → S :=
  fun h k p => ?_
```

1 goal (optional / hardest)

```
h : ((P → Q) → Q) → R
k : R → S
p : P
-- goals:
⊢ S
```

implApplyToChain

```
theorem implApplyToChain {P Q R S : Prop} :
    (((P → Q) → Q) → R) → (R → S) → P → S :=
  fun h k p => k ?_
```

apply `k` — it needs an `R`

```
h : ((P → Q) → Q) → R
k : R → S
p : P
-- goals:
⊢ R
```

implApplyToChain

```
theorem implApplyToChain {P Q R S : Prop} :
    (((P → Q) → Q) → R) → (R → S) → P → S :=
  fun h k p => k (h ?_)
```

apply `h` — it needs a `(P → Q) → Q`

```
h : ((P → Q) → Q) → R
k : R → S
p : P
-- goals:
⊢ (P → Q) → Q
```

implApplyToChain

```
theorem implApplyToChain {P Q R S : Prop} :
    (((P → Q) → Q) → R) → (R → S) → P → S :=
  fun h k p => k (h (fun f => ?_))
```

the goal is a function → introduce `fun f => ?_`

```
f : P → Q
p : P
-- goals:
⊢ Q
```

implApplyToChain

```
theorem implApplyToChain {P Q R S : Prop} :
    (((P → Q) → Q) → R) → (R → S) → P → S :=
  fun h k p => k (h (fun f => f p))
```

f p closes the last goal

```
f p : Q
h (fun f => f p) : R
k (h (fun f => f p)) : S
```

No goals

Rule 7: stuck on `X → Y` (even `(P→Q)→Q`)? Introduce `fun x => ?_`.

Our rulebook

1. `→` groups to the right (`X → Y → Z` = `X → (Y → Z)`); each top-level left arrow is one `fun` argument.
2. A term may be used **more than once**.
3. If you have an `∧` or `∨`, **deconstruct** it (`⟨a,b⟩` / `match`).
4. An `↔` is a pair `⟨mp, mpr⟩`; project with `.mp` / `.mpr`.
5. To **build** an `∧` / `↔`, build each side and combine with `⟨_, _⟩`.
6. `¬X` is `X → False`; a proof of `False` closes any goal.
7. Stuck on `X → Y`? Introduce `fun x => ?_`.